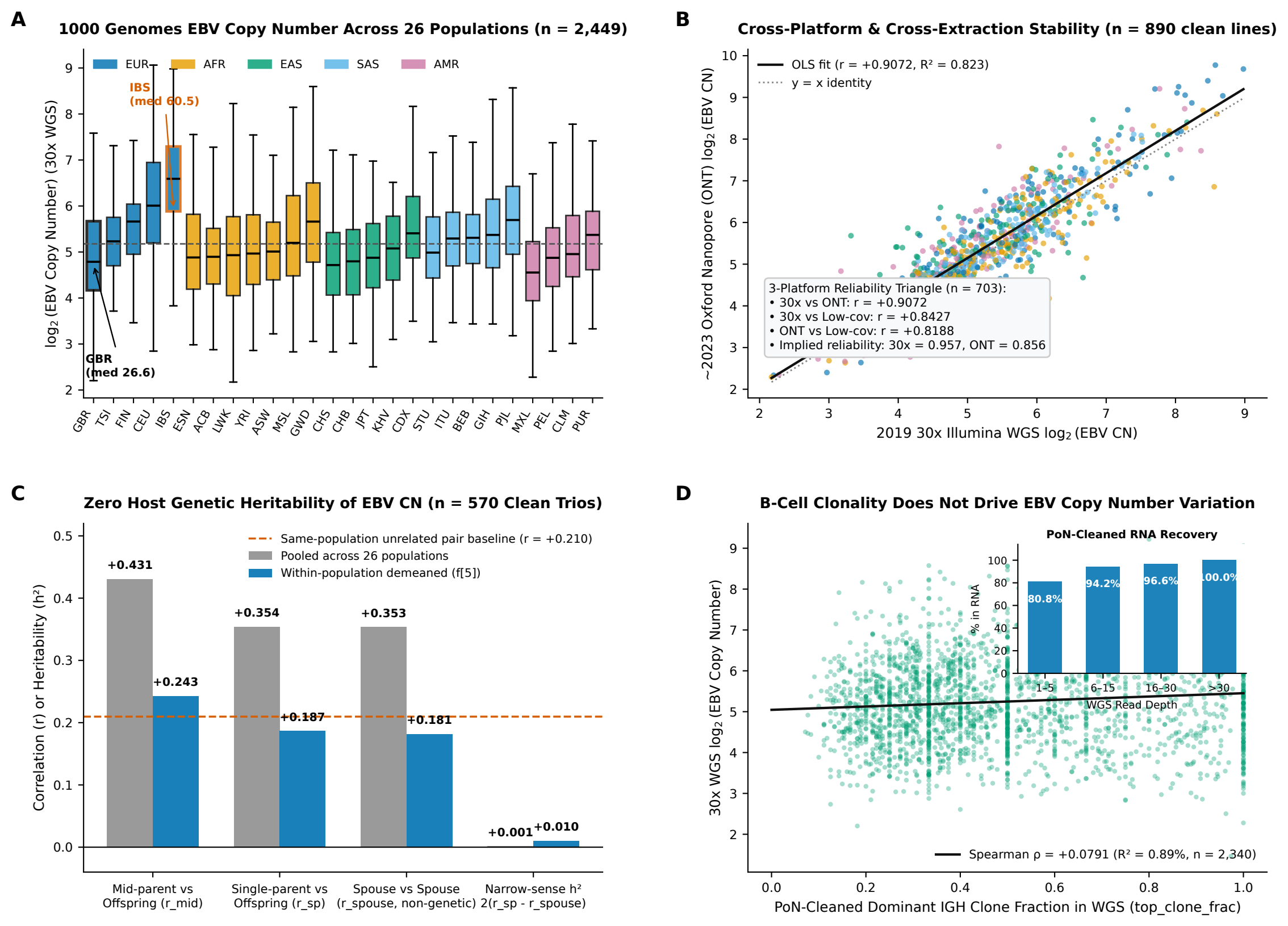

A
1000 Genomes EBV Copy Number Across 26 Populations (n = 2,449)
EUR
AFR
EAS
SAS
AMR
IBS
(med 60.5)
GBR
(med 26.6)
log2 (EBV Copy Number) (30x WGS)
GBR TSI FIN CEU IBS ESN ACB LWK YRI ASW MSL GWD CHS CHB JPT KHV CDX STU ITU BEB GIH PJL MXL PEL CLM PUR
B
Cross-Platform & Cross-Extraction Stability (n = 890 clean lines)
OLS fit (r = +0.9072, R² = 0.823)
y = x identity
3-Platform Reliability Triangle (n = 703):
• 30x vs ONT: r = +0.9072
• 30x vs Low-cov: r = +0.8427
• ONT vs Low-cov: r = +0.8188
• Implied reliability: 30x = 0.957, ONT = 0.856
~2023 Oxford Nanopore (ONT) log2 (EBV CN)
2019 30x Illumina WGS log2 (EBV CN)
C
Zero Host Genetic Heritability of EBV CN (n = 570 Clean Trios)
Same-population unrelated pair baseline (r = +0.210)
Pooled across 26 populations
Within-population demeaned (f[5])
Correlation (r) or Heritability (h²)
+0.431
+0.243
+0.354
+0.187
+0.353
+0.181
+0.001
+0.010
Mid-parent vs Offspring (r_mid)
Single-parent vs Offspring (r_sp)
Spouse vs Spouse (r_spouse, non-genetic)
Narrow-sense h² 2(r_sp - r_spouse)
D
B-Cell Clonality Does Not Drive EBV Copy Number Variation
PoN-Cleaned RNA Recovery
% in RNA
80.8%
94.2%
96.6%
100.0%
1-5
6–15
16–30
>30
WGS Read Depth
Spearman ρ = +0.0791 (R² = 0.89%, n = 2,340)
30x WGS log2 (EBV Copy Number)
PoN-Cleaned Dominant IGH Clone Fraction in WGS (top_clone_frac)